PHP_FLOAT_DIG is the number of digits that can be rounded in a float without precision loss. In other words, the maximum number of digits allowed in the mantissa before we need to worry about rounding errors:

```
$num = 1234567890123456; // 1.234567890123456 x 10 ** 15 fifteen
                         decimal digits
$floatNum = (float) $num;
var_dump($num); // int(1234567890123456)
var_dump($floatNum); // float(1234567890123456)
var_dump((int) $floatNum); // int(1234567890123456)

$num = 12345678901234567; // 1.2345678901234567 x 10 ** 15 sixteen
                          decimal digits
$floatNum = (float) $num;
var_dump($num); // int(12345678901234567)
var_dump($floatNum); // float(12345678901234568)
var_dump((int) $floatNum); // int(12345678901234568) rounding error
                           is present

Output:
int(1234567890123456)
float(1234567890123456)
int(1234567890123456)
int(12345678901234567)
float(12345678901234568)
int(12345678901234568)
```

Let's play a bit more and see what else we can break:

```
$largeFloat = PHP_FLOAT_MAX;

var_dump($largeFloat + 1);
var_dump($largeFloat + 1E291);
var_dump($largeFloat + 1E292);

Output:
float(1.7976931348623157E+308)
float(1.7976931348623157E+308)
float(INF)
```